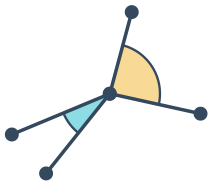
Review: Angle relationships

1 Determine whether the following diagrams show a pair of adjacent angles.

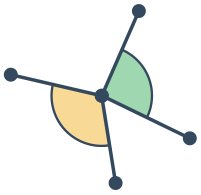
a



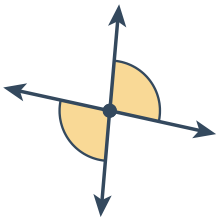
b



c

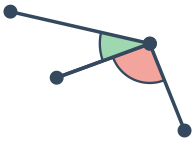


d

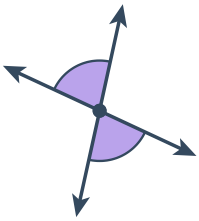


2 Determine whether the following diagrams show a pair of vertical angles.

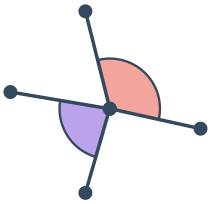
a



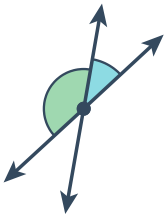
b



c

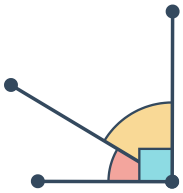


d



3 Determine whether the following diagrams show a pair of complementary angles.

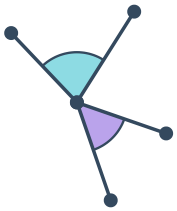
a



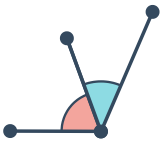
b



c



d

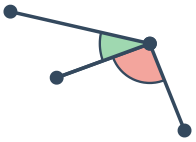


4 Determine whether the following diagrams show a pair of supplementary angles.

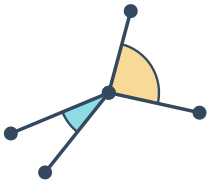
a



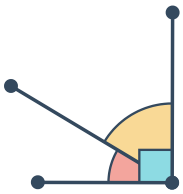
b



c



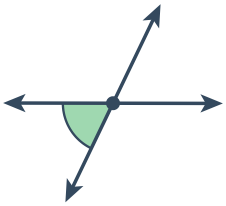
d



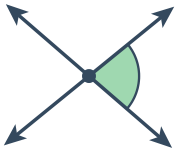
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5 Determine whether the following diagrams show a pair of perpendicular lines.

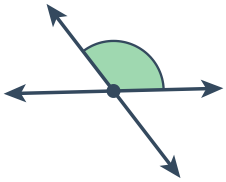
a



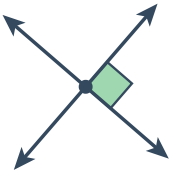
b



c



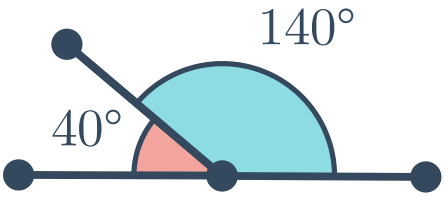
d



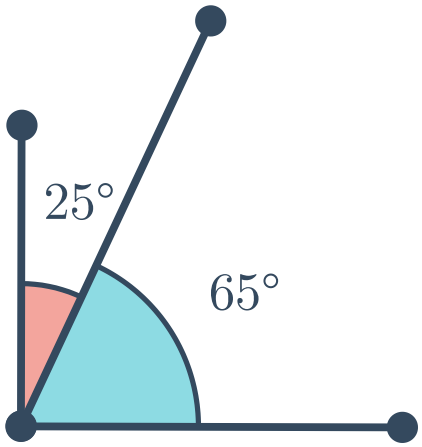
6 Two angles are supplementary. What do their measures add to?

7 Determine whether the angles in each of the following diagrams are complementary.

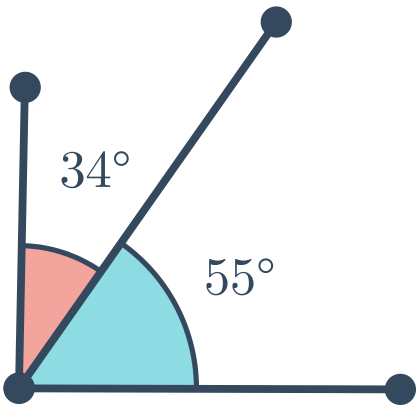
a



b



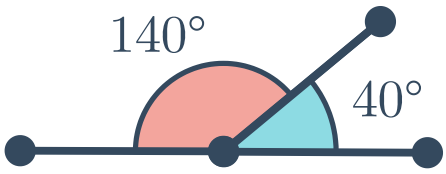
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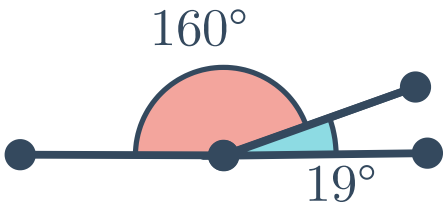
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8 Determine whether the angles in each of the following diagrams are supplementary.

a

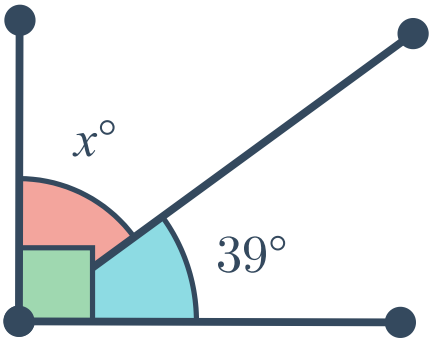


b

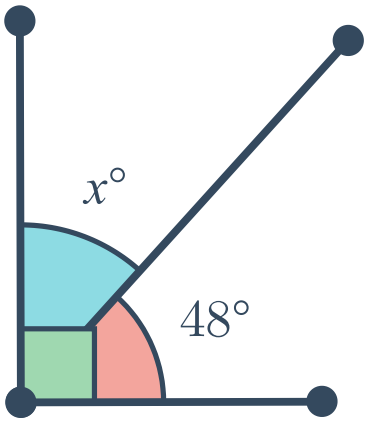


9 Given that the angles are complementary, find the value of x in each of the following diagrams:

a



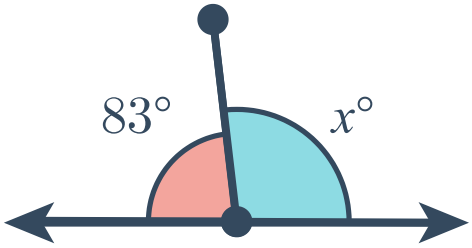
b



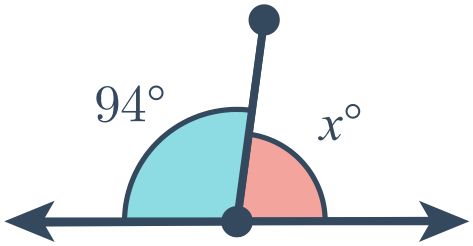
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10 Given that the angles are supplementary, find the value of x in each of the following diagrams:

a

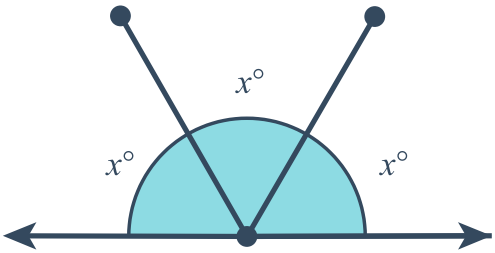


b

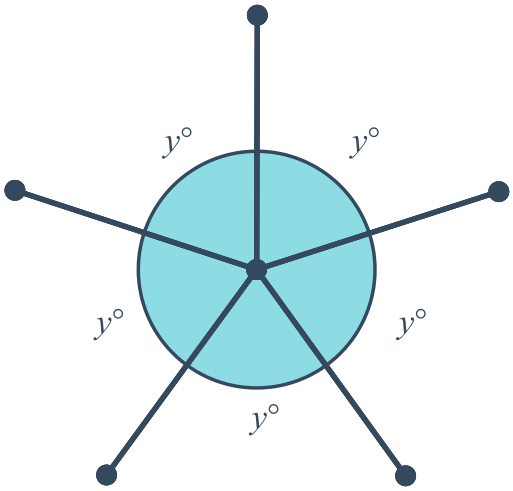


11 Find the value of the marked angle in each of the diagrams below.

a Solve for x.

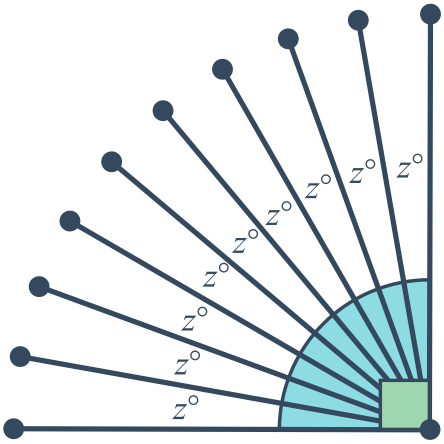


b Solve for y.



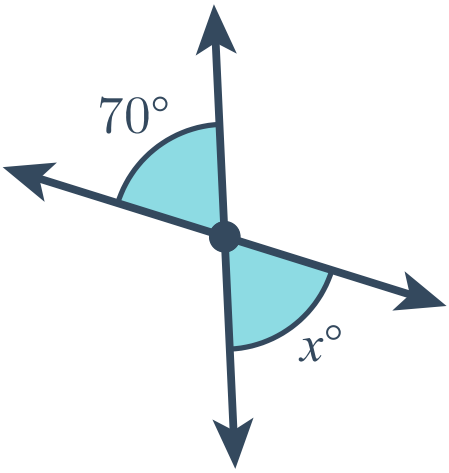
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c Solve for z.

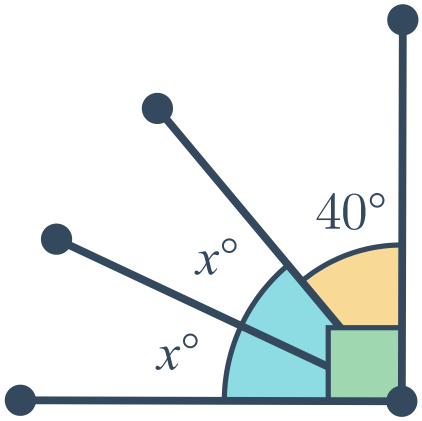


12 Solve for the value of x in the each of the following diagrams:

a

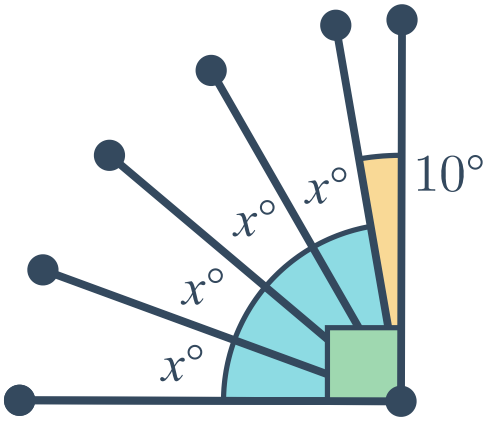


b

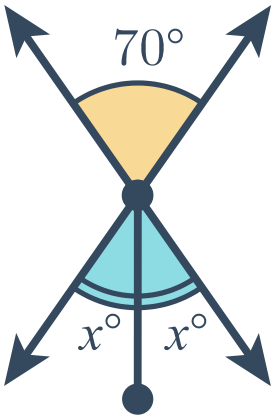


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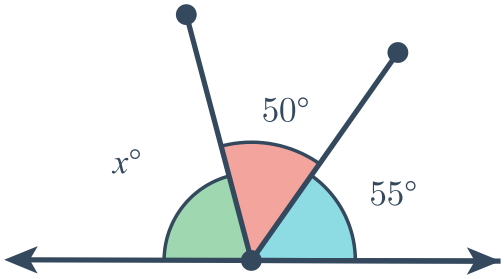
c



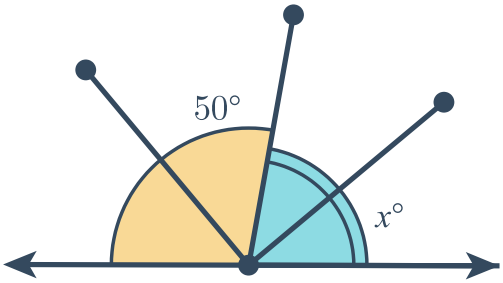
d



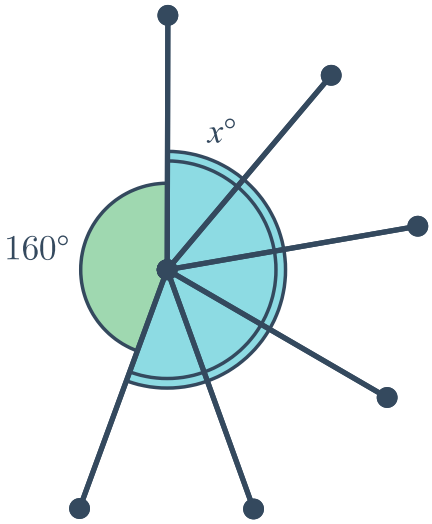
e



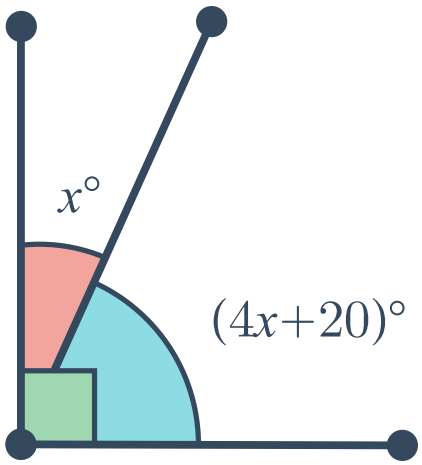
f



g

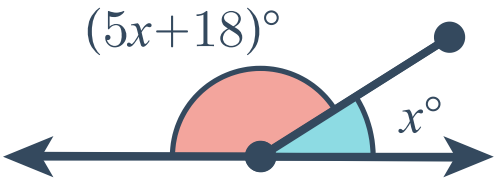


h

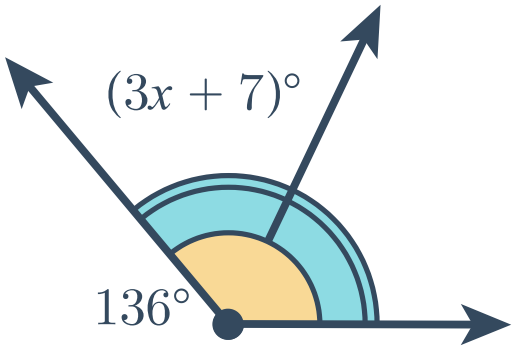


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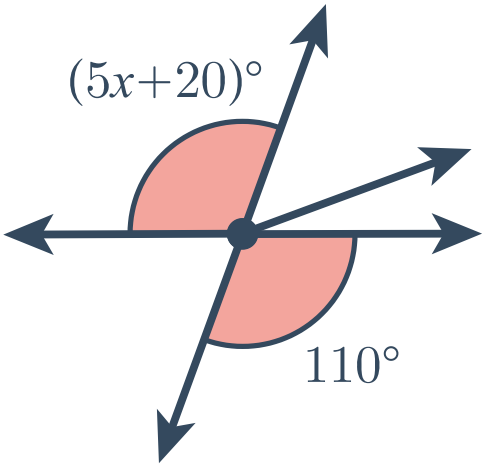
i



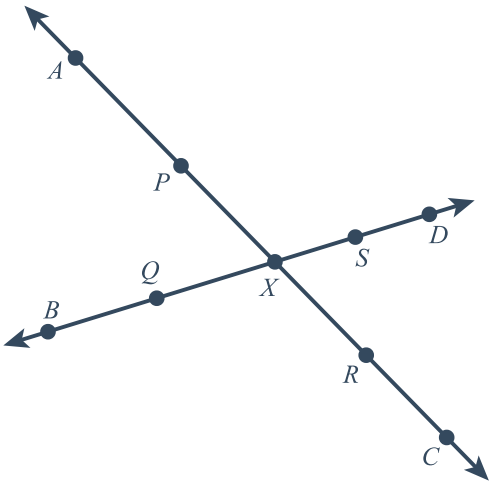
j



k

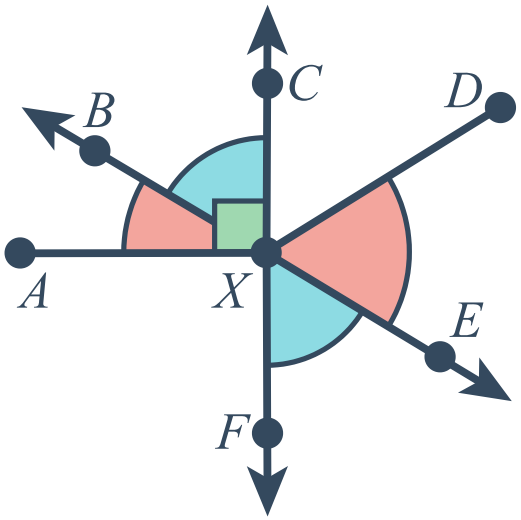


13 Name an angle that is supplementary with $\angle CXD$ in the figure shown.



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14 Consider the diagram below:

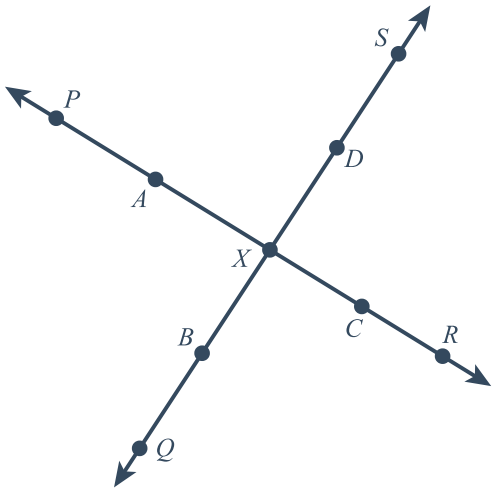


a Identify the two angles that are adjacent.

b Identify the two angles that are vertical.

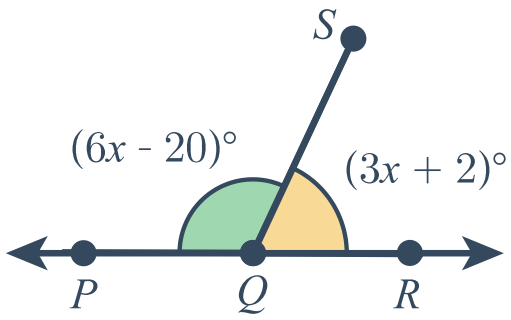
c Identify the two angles that are complementary.

15 Name an angle that is vertical with $\angle AXB$ in the diagram below.



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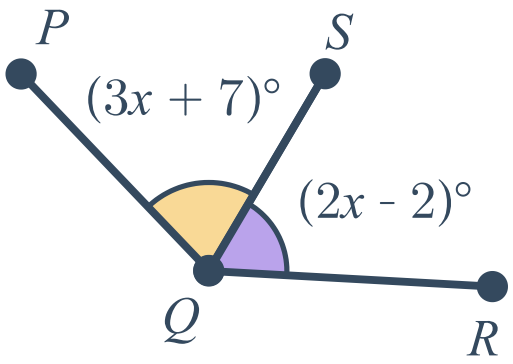
16 Consider the diagram below, where $\angle PQR$ is a straight angle.



a Solve for x.

b Find $m\angle SQR$.

17 Consider the diagram below, where $m\angle PQR = 145^\circ$.

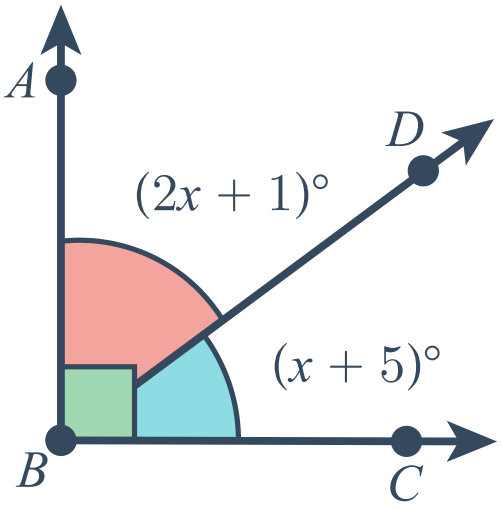


a Solve for x.

b Find $m\angle SQR$.

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18 Consider the angles in the diagram below.



a Solve for x.

b Find $m\angle DBC$.

19 Two angles (in degrees) are supplementary, and one angle is 3 times larger than the other. Let x be the measure of the smaller angle and y be the measure of the larger angle.

a Solve for x.

b Solve for y.

20 Given that two angles form a linear pair, determine whether the following statements are always true.

a They are complementary.

b They are congruent.

c They are adjacent.

d They are supplementary.

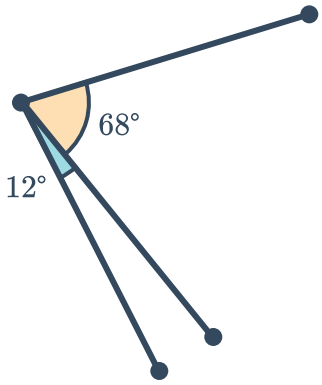
21 Suppose that $\angle MAT$ and $\angle TAP$ form a linear pair, and $m\angle MAT = 148^\circ$. Find $m\angle TAP$.

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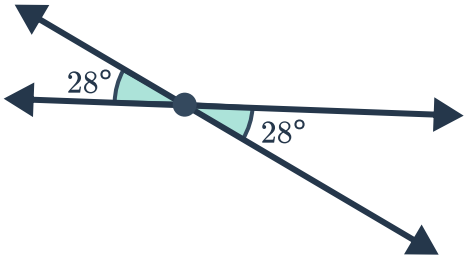
Additional questions

22 Determine whether each of the following diagrams shows the given angle pair. Explain.

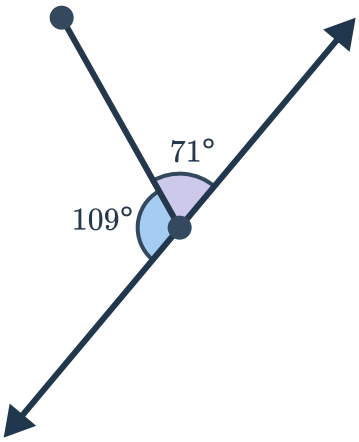
a Complementary angles



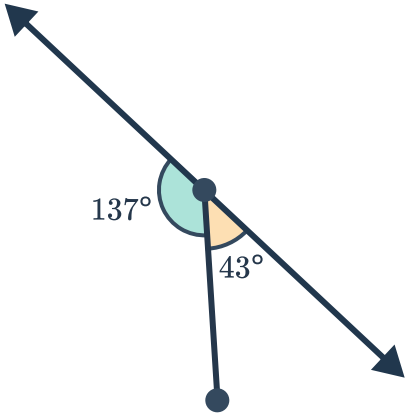
b Vertical angles



c Linear pair



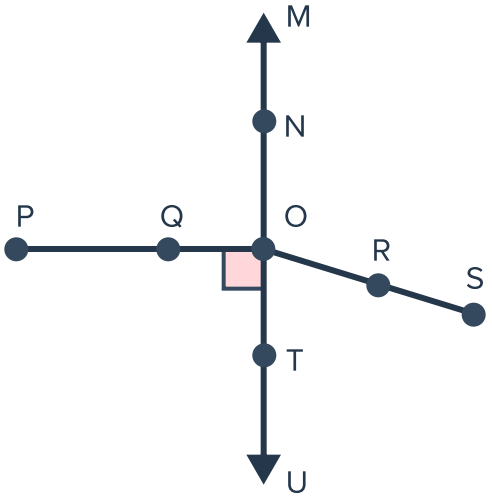
d Linear pair



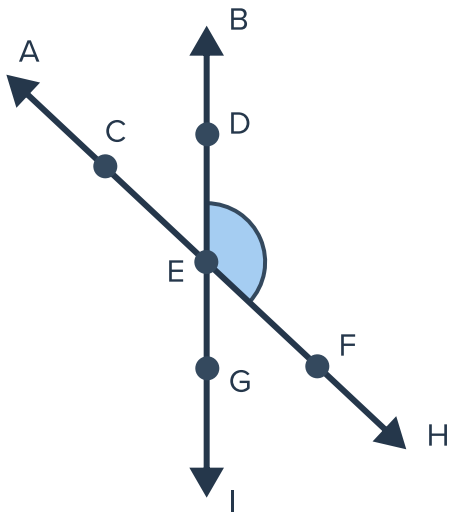
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23 Identify the missing angle for the angle pair.

a ▯ forms a linear pair with $\angle QOT$.

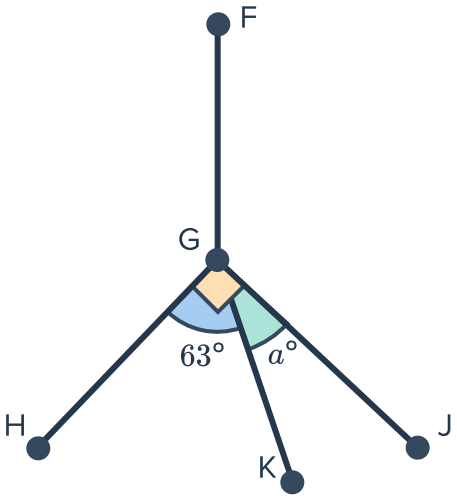


b ▯ forms a vertical pair with $\angle BEH$.

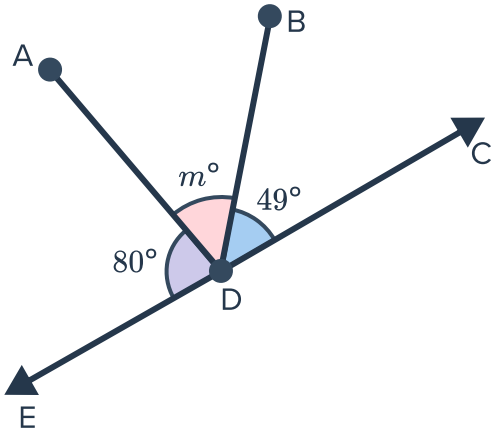


24 Determine the value of the variable in each of the following diagrams.

a

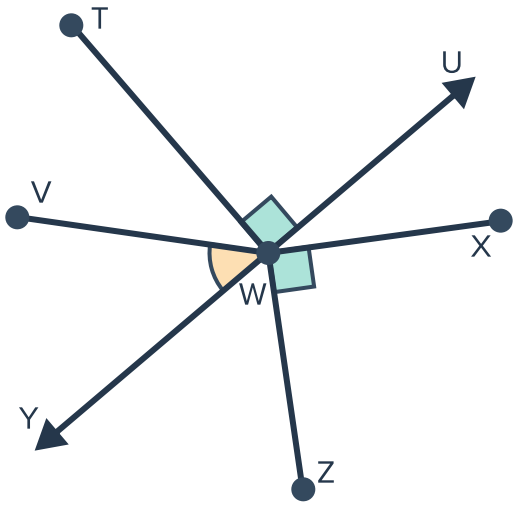


b



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25 Use the diagram to find an example of each angle pair if it exists.



a Linear pair

b Vertical angles

c Complementary angles

d Supplementary angles that are not also a linear pair

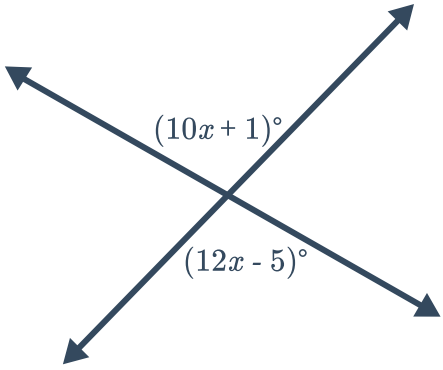
26 For each diagram:

i Write an equation that models the relationship shown in the diagram.

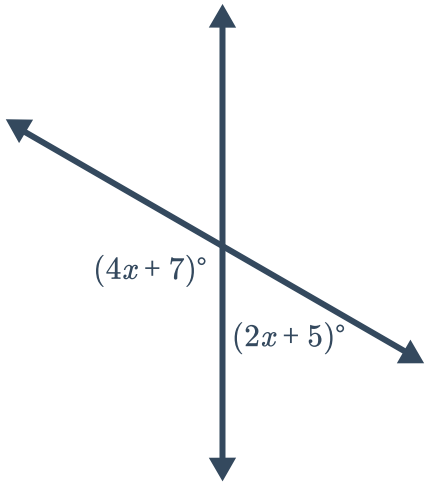
ii What theorem or postulate justifies this equation?

iii Solve for x.

a



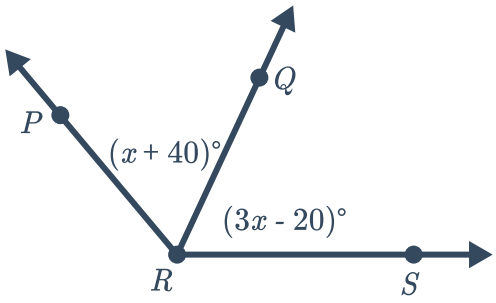
b



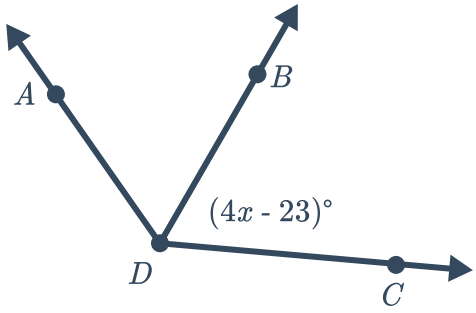
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27 Use the given information to solve for the value of x.

a $\overrightarrow{RQ}$ bisects $\angle PRS$



b $\overrightarrow{DB}$ bisects $\angle ADC$ and $m\angle ADC = 138^\circ$



28 The measure of an angle's supplement is three times the measure of the angle. What is the measure of the angle?

29 The measure of an angle's complement is five less than half the measure of the angle's supplement. What is the measure of the angle?

30 Given:

- $m\angle 1 = 2x - 30$
- $m\angle 2 = 5x - 120$
- $m\angle 3 = 10x - 150$
- $\angle 1$ and $\angle 3$ form a linear pair
- $\angle 1$ and $\angle 2$ are vertical angles

Find:

a $m\angle 1$

b $m\angle 3$

31 Determine if the statement is always, sometimes, or never true. Create diagrams to justify your answers.

a Complementary angles are also adjacent.

b A linear pair is supplementary.

c An obtuse angle forms a linear pair with an acute angle.

d Vertical angles form a linear pair.

e Vertical angles are supplementary.

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32 Determine if the information given is enough to justify the conclusion. Draw a diagram to support your solution.

a Given: $\angle ABC \cong \angle DBF$
Conclusion: $\angle ABC$ and $\angle DBF$ are vertical angles

b Given: $\angle MNP$ and $\angle PNR$ form a linear pair and $\overline{NP}$ bisects $\angle MNR$.
Conclusion: $m\angle MNP = m\angle PNR = 90^\circ$

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